

WILLIAMS 35 NNW, AZ, USA

WMO: 722770

Lat: 35.7552N

Lon: 112.3374W

Elev: 5990

StdP: 11.78

Time Zone: -7.00 (W07)

Period: 08-19

WBAN: 53155

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	10.7	15.4	-8.0	4.5	41.3	-3.2	5.8	40.6	21.0	36.1	18.7	36.5	3.1	N/A	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.1	91.2	55.7	89.1	55.8	86.9	55.5	62.4	75.9	61.4	75.7	60.4	75.0	8.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.8	92.7	65.9	57.3	87.7	65.7	55.9	83.2	65.6	31.3	75.8	30.5	75.2	29.7	75.1	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	18.1	15.7		0.4	95.4	5.8	1.9	-3.8	96.8	-7.1	97.9	-10.4	98.9	-14.6	100.3
				0.2	64.6	5.8	1.0	-4.0	65.3	-7.4	65.9	-10.6	66.4	-14.8	67.1
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	34.6	36.9	44.1	49.1	56.0	68.3	72.2	70.0	64.3	53.1	43.2	33.6	(6)
	DBStd	15.02	7.63	8.61	6.63	7.09	7.16	5.37	3.24	3.44	4.81	6.73	7.53	8.50	(7)
	HDD50	1950	477	367	200	99	31	0	0	0	1	48	219	508	(8)
	HDD65	5233	942	787	649	477	284	28	0	6	66	368	653	973	(9)
	CDD50	2755	0	1	16	72	218	548	689	620	430	145	16	0	(10)
	CDD65	564	0	0	0	0	6	127	224	161	45	1	0	0	(11)
	CDH74	8228	0	0	0	29	329	2566	2512	1905	834	53	0	0	(12)
	CDH80	3042	0	0	0	1	47	1102	1039	677	174	2	0	0	(13)
(14) Wind	WSAvg	5.9	5.0	5.9	6.7	7.5	7.0	6.6	5.4	5.0	5.2	5.4	5.7	5.5	(14)
(15) Precipitation	PrecAvg	13.6	1.1	1.1	1.0	0.8	0.4	0.4	1.8	2.2	1.4	1.2	0.7	1.2	(15)
	PrecMax	20.7	4.6	3.7	2.1	1.7	1.7	2.0	3.4	4.5	3.6	4.3	2.9	2.9	(16)
	PrecMin	7.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.1	0.0	0.0	0.1	(17)
	PrecStd	3.8	1.2	0.8	0.7	0.5	0.5	0.5	0.9	1.1	1.0	1.4	0.7	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	66.4	71.9	77.4	84.5	94.9	93.5	91.2	87.6	79.0	71.0	60.4	(19)
		MCWB	40.1	42.9	46.3	48.3	50.8	55.9	57.8	56.1	55.2	51.0	46.5	41.3	(20)
	2%	DB	55.1	62.4	67.7	73.8	81.0	92.1	90.7	88.8	84.4	75.1	66.7	56.0	(21)
		MCWB	38.7	41.8	44.2	47.0	50.0	54.8	57.2	57.1	55.1	48.5	44.0	39.2	(22)
	5%	DB	51.8	57.6	64.2	70.5	77.9	89.5	88.8	86.7	81.9	72.5	63.0	52.0	(23)
		MCWB	37.4	40.4	42.8	45.6	48.4	53.8	57.1	57.4	54.4	48.5	42.8	37.8	(24)
	10%	DB	47.4	52.1	60.4	67.2	74.6	86.8	86.5	84.2	79.2	69.3	59.1	47.3	(25)
		MCWB	35.6	38.1	41.3	44.1	47.6	53.0	56.9	57.6	54.1	47.6	41.7	36.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	46.1	47.1	50.2	52.4	59.3	64.0	63.6	61.2	55.5	47.9	44.4	(27)
		MCDB	52.3	58.5	69.2	74.0	78.3	82.7	76.8	76.3	73.9	67.1	66.1	52.6	(28)
	2%	WB	40.9	42.9	45.4	48.2	50.9	57.4	62.8	62.4	59.7	53.1	45.8	42.2	(29)
		MCDB	50.8	58.6	65.8	71.1	76.5	84.8	76.0	75.8	73.2	66.5	62.3	51.1	(30)
	5%	WB	39.1	41.3	43.8	46.5	49.5	56.0	61.9	61.4	58.4	51.2	43.9	40.1	(31)
		MCDB	48.1	55.8	62.7	68.2	74.4	84.0	76.1	75.4	72.6	66.1	59.4	48.7	(32)
	10%	WB	37.3	39.5	42.0	45.0	48.1	54.5	60.8	60.4	57.1	49.5	42.2	37.8	(33)
		MCDB	47.0	52.4	58.9	65.9	71.9	82.5	75.4	75.1	71.8	64.3	56.6	47.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	23.1	26.8	28.5	28.9	33.0	27.1	26.8	27.8	27.3	25.7	20.5	(35)
		MCDBR	27.3	31.8	32.7	34.0	33.7	36.0	31.4	31.2	31.4	31.9	31.0	27.6	(36)
	5% WB	MCWBR	15.8	16.9	15.6	15.9	14.8	14.2	10.5	10.4	12.2	14.4	15.8	15.5	(37)
		MCWBR	23.2	28.7	29.9	31.2	31.6	31.9	22.9	23.5	24.5	24.3	27.7	22.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.241	0.246	0.262	0.278	0.291	0.295	0.344	0.351	0.297	0.278	0.246	0.239	(40)
		taud	2.635	2.631	2.577	2.517	2.530	2.504	2.446	2.414	2.576	2.581	2.633	2.643	(41)
	Ebn at Noon		314	323	323	319	314	311	295	292	306	305	309	308	(42)
		Edh at Noon	22	25	29	33	33	34	36	36	29	26	22	21	(43)
(44) All-Sky Solar Radiation	RadAvg		984	1268	1752	2190	2448	2646	2152	1996	1862	1467	1111	870	(44)
	RadStd		85	117	122	110	181	142	149	116	84	113	52	66	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	+133	+47

Nomenclature: See separate page